

TELCO IBM CUSTOMER CHURN ANALYSIS REPORT

Python | SQL | Excel | Power BI

Data Analyst

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Executive Summary

This report details the comprehensive analysis of customer churn undertaken to identify key drivers and provide actionable insights. By leveraging SQL for data extraction and manipulation, Power BI for visualization, and Excel for supplementary analysis, we have gained a deep understanding of why customers are leaving. The findings presented herein will guide strategic decision-making to improve customer retention and drive sustainable business growth.

Project Overview

The primary goal of this project was to conduct a thorough analysis of customer churn patterns. This involved exploring historical customer data, identifying significant factors contributing to churn, and developing a predictive understanding of at-risk customers. The insights derived are intended to empower the business with data-driven strategies for customer retention.

Business Problem

High customer churn rates present a significant challenge to business growth and profitability. Understanding the underlying reasons for customer attrition is crucial for developing effective retention strategies. Without a clear understanding of churn drivers, the business risks continued revenue loss and a decline in market share.

Business Objectives

The key objectives of this analysis were to:

- Identify the primary drivers of customer churn.
- Quantify the impact of various factors on churn probability.
- Provide data-backed recommendations for improving customer retention.
- Establish a foundation for ongoing churn monitoring and prediction.
- Develop a clear and actionable understanding of customers leading to churn.

Dataset Overview

The analysis was performed on a comprehensive dataset containing Customer Profile, Core Services, Additional Services, Streaming Services, Contract, Billing, Revenue, number of Customers Churned and Retained. The dataset comprises 7032 records and 8 main features, providing a rich source of information for understanding customer behaviour.

Key Data Attributes:

- Customer Churn Rate
- Contract Type
- Tenure
- Phone Service
- Multiple Lines Distribution
- Internet Service
- Additional Services
- Streaming Services
- Paperless Billing
- Payment Methods

Tools & Technologies

A combination of industry-standard tools was employed to ensure a robust and efficient analysis:

- **Excel:** Employed for supplementary data analysis, and detailed examination of specific data segments.
- **Python:** Data cleaning, pre-processing, handling missing values, correcting data types, data validation, and preparing the dataset for analysis
- **SQL:** Utilized for Exploratory Data Analysis (EDA), aggregation, filtering, grouping, and answering business questions through queries
- **Power BI:** Leveraged for creating interactive and insightful dashboards, enabling visualization of churn trends and key performance indicators.

Project Methodology

The project followed a structured methodology:

1. **Data Extraction & Understanding:** Retrieved relevant data using SQL and performed an initial exploration to understand data structure and quality.
2. **Data Cleaning & Pre-processing:** Handled missing values, corrected inconsistencies, and transformed data into a suitable format for analysis.
3. **Exploratory Data Analysis (EDA):** Analysed key variables and their relationships with churn using statistical methods and visualizations.
4. **Feature Engineering:** Created new features where necessary to enhance the predictive power of the analysis.
5. **Churn Driver Identification:** Identified and quantified the factors most strongly associated with customer churn.
6. **Visualization & Reporting:** Developed Power BI dashboards to present findings and created this comprehensive report.

Dashboard Overview

The accompanying Power BI dashboard provides a dynamic and interactive view of customer churn. Key features include:

- **Overall Churn Rate:** A clear indicator of the current churn situation.
- **Churn by Demographics:** Breakdown of churn across different age groups, genders, and locations.
- **Service Usage Impact:** Visualization of how usage patterns affect churn.
- **Contractual Factors:** Analysis of churn based on contract type, tenure, and charges.
- **Customer Support Trends:** Insights into the relationship between support interactions and churn.
- **Key Churn Drivers:** Highlights of the most influential factors contributing to churn

Telco Customers Churn Analysis

Yes - Churned No - Retained

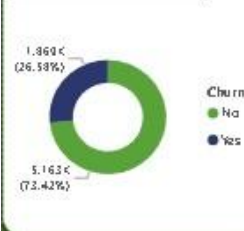
Yes
1.869K

No
5.163K

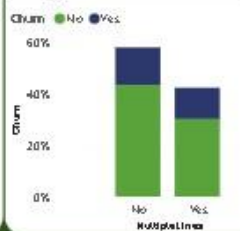
Internet Se...
☐ DSL
☐ Fiber optic
☐ No

Contract
☐ Month-to-month
☐ One year
☐ Two year

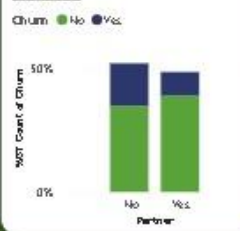
Customer Churn Rate



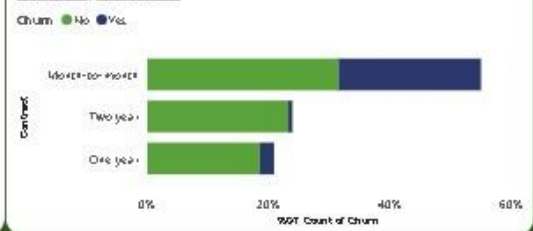
MultipleLines Distribution



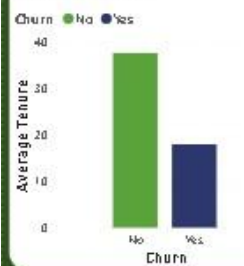
Partner



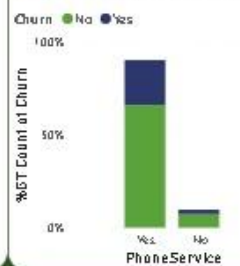
Churn by Contract



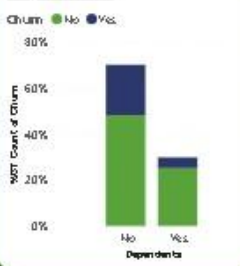
Average Tenure



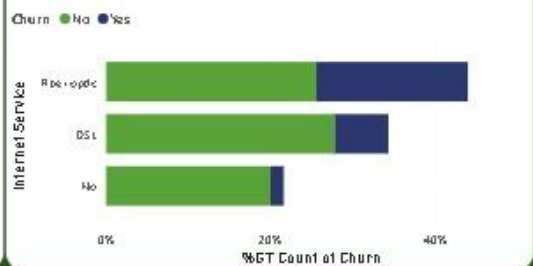
PhoneService Distribution



Dependents



InternetService Distribution



Yes - Churned No - Retained

Senior Citizen

☐ 0
☐ 1

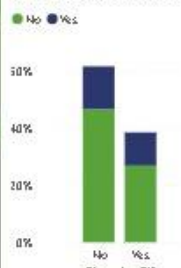
Gender

☐ Female
☐ Male

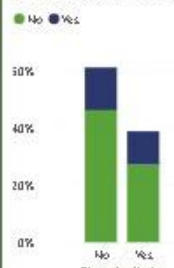
MonthlyCharges

18.25 118.75

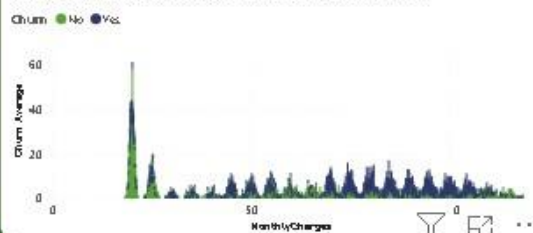
StreamingServices



Streaming Movies



Average Monthly Charges affecting Churn Rate



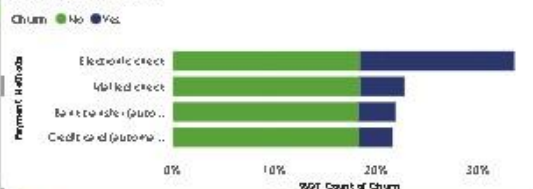
Gender



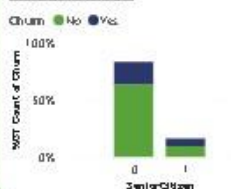
Paperless Billing



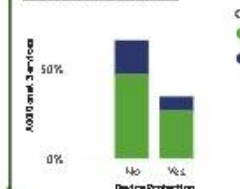
Payment Method



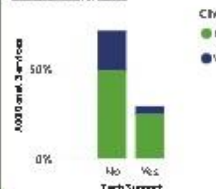
Senior Citizen



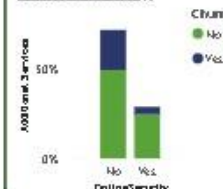
Device Protection



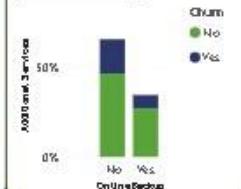
Tech Support



Online Security



Online Backup



Detailed Analysis

9.1 Customer Churn Rate

Observation

There are 7,032 customers, of which 1,869 customers have churned and 5,163 customers have been retained.

Interpretation

The company has an overall customer retention rate of 73.42% and a customer churn rate of 26.58%, indicating that approximately one out of every four customers leaves the company.

Recommendation

The company should investigate all major business areas to identify the factors contributing most to customer churn and prioritize retention strategies accordingly.

9.2 Multiple Lines Distribution

Observation

Customers with Multiple Lines

- Customers Churned: 850 (12.09%)
- Customers Retained: 2,117 (30.11%)

Customers without Multiple Lines

- Customers Churned: 1,019 (14.49%)
- Customers Retained: 3,046 (43.32%)

Interpretation

Customers without Multiple Lines account for a slightly larger proportion of Customers churned. The difference between the two groups is approximately 2.4%

Recommendation

Promote Multiple Line plans to customer segments that would benefit from them, such as businesses and professionals with higher communication requirements.

9.3 Contract Type

Observation

Month-to-Month Contract

- Customers Churned: 1,655 (23.54%)
- Customers Retained: 2,220 (31.57%)

One-Year Contract

- Customers Churned: 166 (2.36%)
- Customers Retained: 1,306 (18.57%)

Two-Year Contract

- Customers Churned: 48 (0.68%)
- Customers Retained: 1,637 (23.28%)

Interpretation

Customers with Month-to-Month contracts account for the highest proportion of churned customers. In comparison, customers with One-Year and Two-Year contracts show significantly lower churn, indicating that longer-term contracts are associated with better customer retention.

Recommendation

The company should prioritize Month-to-Month customers by providing excellent onboarding, proactive customer support, personalized offers, and incentives to encourage them to switch to longer-term contracts.

9.4 Internet Service Distribution

Observation

Fiber Optic

- Customers Churned: 1,297 (18.44%)
- Customers Retained: 1,799 (25.58%)

DSL

- Customers Churned: 459 (6.53%)
- Customers Retained: 1,957 (27.83%)

No Internet Service

- Customers Churned: 113 (1.61%)
- Customers Retained: 1,407 (20.01%)

Interpretation

Fiber Optic customers represent the largest proportion of churned customers, making Internet Service one of the strongest factors associated with customer churn.

Recommendation

The company should investigate why Fiber Optic customers are leaving by evaluating service quality, pricing, customer support, and customer satisfaction. Expansion of Fiber Optic services should continue while ensuring service quality meets customer expectations.

9.5 Customer Tenure

Observation

- Average Tenure of Retained Customers: 37.65 months
- Average Tenure of Churned Customers: 17.98 months

Interpretation

Customers who churn have, on average, less than half the tenure of retained customers. This indicates that newer customers are more likely to leave than long-term customers.

Recommendation

The company should strengthen customer engagement during the early months by providing loyalty programs, onboarding assistance, exclusive offers, and personalized communication to improve long-term retention.

9.6 Additional Services

Observation

Customers without additional services consistently account for more churned customers.

Online Backup

Without Online Backup:

- Customers Churned: 1,346 (19.14%)
- Customers Retained: 3,261 (46.37%)

With Online Backup:

- Customers Churned: 523 (7.44%)
- Customers Retained: 1,902 (27.05%)

Online Security

Without Online Security:

- Customers Churned: 1,574 (22.38%)
- Customers Retained: 3,443 (48.96%)

With Online Security:

- Customers Churned: 295 (4.20%)
- Customers Retained: 1,720 (24.46%)

Device Protection

Without Device Protection:

- Customers Churned: 1,324 (18.83%)
- Customers Retained: 3,290 (46.79%)

With Device Protection:

- Customers Churned: 545 (7.75%)
- Customers Retained: 1,873 (26.64%)

Tech Support

Without Tech Support:

- Customers Churned: 1,559 (22.17%)
- Customers Retained: 3,433 (48.82%)

With Tech Support:

- Customers Churned: 310 (4.41%)
- Customers Retained: 1,730 (24.60%)

Interpretation

Across all additional services, customers who do not subscribe to these services consistently represent a larger proportion of churned customers. This suggests that customers using additional services tend to remain with the company longer.

Recommendation

The company should increase awareness and adoption of additional services through bundled plans, promotional offers, and personalized recommendations where appropriate.

9.7 Streaming Services

Observation

Streaming TV

Without Streaming TV:

- Customers Churned: 1,055 (15.00%)
- Customers Retained: 3,274 (46.56%)

With Streaming TV:

- Customers Churned: 814 (11.58%)
- Customers Retained: 1,889 (26.86%)

Streaming Movies

Without Streaming Movies:

- Customers Churned: 1,051 (14.95%)
- Customers Retained: 3,250 (46.22%)

With Streaming Movies:

- Customers Churned: 818 (11.63%)
- Customers Retained: 1,913 (27.20%)

Interpretation

Streaming TV and Streaming Movies follow nearly identical patterns. Customers without these services account for a slightly larger proportion of churned customers, although the difference is relatively small.

Recommendation

No immediate business action is recommended based solely on these variables. They should continue to be monitored alongside other customer behavior indicators.

9.8 Phone Service Distribution

Observation

Without Phone Service

- Customers Churned: 170 (2.42%)
- Customers Retained: 510 (7.25%)

With Phone Service

- Customers Churned: 1,699 (24.16%)
- Customers Retained: 4,653 (66.17%)

Interpretation

Customers with Phone Service make up the majority of the customer base and therefore account for both the highest retention and the highest number of churned customers. Overall, Phone Service does not appear to be a primary driver of churn.

Recommendation

The company should continue improving Phone Service quality and customer satisfaction but prioritize retention efforts toward factors showing stronger associations with churn.

9.9 Paperless Billing

Observation

Without Paperless Billing:

- Customers Churned: 469 (6.67%)
- Customers Retained: 2,395 (34.06%)

With Paperless Billing:

- Customers Churned: 1,400 (19.98%)
- Customers Retained: 2,768 (39.36%)

Interpretation

Customers using Paperless Billing show both high retention and high churn because a larger proportion of customers use this billing method. The data alone does not indicate that Paperless Billing causes churn.

Recommendation

The company should continue promoting Paperless Billing because it improves operational efficiency while further investigating whether any customer experience issues exist within the billing process.

9.10 Payment Methods

Observation

Electronic Check

- Customers Churned: 1,071 (15.23%)
- Customers Retained: 1,294 (18.40%)

Mailed Check

- Customers Churned: 308 (4.38%)
- Customers Retained: 1,296 (18.43%)

Bank Transfer (Automatic)

- Customers Churned: 258 (3.67%)
- Customers Retained: 1,284 (18.26%)

Credit Card (Automatic)

- Customers Churned: 232 (3.30%)
- Customers Retained: 1,289 (18.33%)

Interpretation

Electronic Check users account for the largest proportion of churned customers compared with other payment methods, while automatic payment methods show comparatively lower churn.

Recommendation

The company should investigate the customer experience associated with Electronic Check payments and encourage customers to adopt automatic payment methods by highlighting their convenience and reliability.

Overall Business Insights

The analysis reveals that customer churn is a multifaceted issue driven by a combination of factors including contractual terms, service value perception, customer support quality, and initial customer experience. Customers on flexible, month-to-month contracts are most at risk, especially within their first year of service. High charges without perceived value, low service engagement, and poor support experiences are significant contributors to dissatisfaction and eventual churn. Conversely, customers who utilize value-added services like online security and backup, and those with longer-term commitments, demonstrate higher loyalty.

Business Recommendations

Based on the analysis, the following strategic recommendations are proposed to mitigate customer churn:

1. **Enhance Early Customer Experience:** Implement robust onboarding and proactive customer success initiatives for new customers, focusing on the first 12 months.
2. **Incentivize Long-Term Contracts:** Offer attractive discounts or benefits for customers who commit to 1-year or 2-year contracts.
3. **Optimize Pricing and Value Communication:** Clearly articulate the value proposition for all service tiers, especially higher-priced ones. Consider personalized offers for at-risk high-value customers.
4. **Boost Customer Engagement:** Develop campaigns to encourage deeper and more consistent usage of core services. Highlight new features and benefits regularly.
5. **Improve Customer Support:** Invest in training, streamline processes, and focus on first-contact resolution to enhance customer satisfaction with support interactions.
6. **Promote Value-Added Services:** Increase awareness and adoption of services like online security and backup, highlighting their contribution to overall value and security.
7. **Segmented Marketing:** Tailor communication and offers based on customer demographics and behavior patterns identified as high-churn risks.

Key Learnings

This project has underscored several key learnings:

- **The critical importance of the initial customer journey.** First impressions and early support significantly influence long-term retention.
- **Value perception is paramount.** Customers will churn if they do not believe they are receiving adequate value for their money, regardless of contract length.
- **Proactive engagement is more effective than reactive problem-solving.** Identifying and addressing potential issues before they lead to churn is crucial.
- **Data-driven insights are essential for strategic decision-making.** A clear understanding of churn drivers enables targeted and efficient retention efforts.
- **The integrated use of SQL and Power BI is powerful** for comprehensive data analysis and insightful visualization.

Future Improvements

To further enhance churn prediction and prevention efforts, consider the following:

- **Develop a predictive churn model:** Utilize machine learning techniques to predict the likelihood of individual customers churning.
- **Real-time Churn Monitoring:** Implement systems for continuous monitoring of key churn indicators.
- **A/B Testing Retention Strategies:** Systematically test the effectiveness of different retention offers and interventions.
- **Incorporate Customer Feedback:** Integrate customer feedback data (surveys, reviews) into the churn analysis.
- **Analyze Competitor Offerings:** Conduct regular benchmarking against competitor products, pricing, and service quality.

Conclusion

This Customer Churn Analysis Report provides a clear roadmap for understanding and addressing customer attrition. By implementing the recommended strategies, the business can significantly improve customer retention, foster loyalty, and achieve sustainable growth. Continuous monitoring and adaptation based on ongoing data analysis will be key to long-term success in managing customer churn.